

Appendix : A Comparative Explanation of Causal Inference Terminology in Medicine and End to End Autonomous Driving (AD) Models

Intervention

* Medicine ; causal inference for treatment effects

Surgery, medication, therapeutic procedure

* AD Models ; causal inference for action (**task**) selection

The intervention corresponds to the action selected by the autonomous driving model.
(acceleration, braking and stopping, right or left steering)

Confounding

* Medicine

Background differences distort treatment effect estimates

(severe patients receive aggressive therapy)

* AD Models

Environmental factors influence both action selection and outcomes

(On narrow roads, night-time conditions combined with low- μ road surfaces increase both braking frequency and accident risk.)

Control Group; a group in which background conditions are held constant (**controlled**) by withholding the intervention

* Medicine

Patients who did not receive the treatment

* AD Models

The counterfactual actions (tasks) that the model did not select;

“what if it had not braked?” or “what if it had gone straight instead of turning?”

In contrast, **Treatment Group**

* Medicine

Patients who actually received the treatment

* AD Models

The set of actions (**tasks**) the model actually selected

(executed a hard brake, turned right or left, stopped)

Covariates

* Medicine

Background factors influencing treatment choice and outcomes

(age, comorbidities, severity, lifestyle, family medical history)

* AD Models

Environmental and vehicle factors influencing action (**task**) choice and outcomes

(weather, night-time, congestion, pedestrian density, ODD conditions, sensor state)

Propensity Score

* Medicine

Probability that a patient receives a treatment given covariates

* AD Models

Probability that the model selects a specific action (**task**) given the observed situation

It is implicitly encoded within the model's internal decision-making structure.

In other words, it is implicitly encoded within the model's internal **purpose** hierarchy.

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A **covariate** is a baseline clinical factor that causally affects both treatment selection and patient outcomes and therefore must be adjusted for as a confounder that opens a backdoor path.

In the context of autonomous driving, the counterparts to **covariates** are baseline environmental and vehicle-state factors that causally influence both the selection of intervention **Tasks** (such as braking or steering) and accident risk, and therefore function as confounders that open backdoor paths and must be statistically adjusted for.

A **propensity score** is the probability that a patient with given baseline **covariates** will receive a particular treatment.

1. More severe patients are more likely to receive aggressive treatments.
2. Younger patients are more likely to receive invasive therapies.
3. Older patients are more likely to receive conservative rather than surgical treatments.

In autonomous driving, a **propensity score** $P_{Se}(x) = P(\text{intervention} \mid X = x)$ represents the probability that a vehicle will select a particular intervention action (**Task**) given observed covariates such as weather, night-time conditions, traffic congestion, pedestrian density, ODD constraints, and sensor state.

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